

***Lab 8***

***Implementation of 2D Transformation & Scaling***

***Course Title: Computer Graphics***

***Course Code: CSE 420***

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***Section: 01***

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## Lab Task

```
#include <iostream>
#include <GL/gl.h>
#include <GL/glut.h>

using namespace std;

// Draw 2D square
void drawSquare() {
    glBegin(GL_QUADS);
    glVertex2f(-0.2f, -0.2f);
    glVertex2f(0.2f, -0.2f);
    glVertex2f(0.2f, 0.2f);
    glVertex2f(-0.2f, 0.2f);
    glEnd();
}

// Draw text
void drawText(const char* text, float r, float g, float b) {
    glColor3f(r, g, b);
    glRasterPos2f(-0.9f, -0.9f);
    while (*text) {
        glutBitmapCharacter(GLUT_BITMAP_HELVETICA_18, *text);
        text++;
    }
}

// Translation
void translate2D() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Original
    glColor3f(1,1,1);
    drawSquare();

    // Translated
    glPushMatrix();
    glTranslatef(0.5f, 0.5f, 1.0f);
    glColor3f(1,0.5,0);
    drawSquare();
    glPopMatrix();

    drawText("Translation 2D",1,0.5,0);
    glFlush();
}

// Scaling
void scale2D() {
    glClear(GL_COLOR_BUFFER_BIT);

    glColor3f(1,1,1);
    drawSquare();
}
```

```

    glPushMatrix();
    glTranslatef(0.5f, 0.5f, 1.0f);
    glScalef(0.9f, 0.3f, 1);
    glColor3f(0,0,1);
    drawSquare();
    glPopMatrix();

    drawText("Scaling 2D",0,0,1);
    glFlush();
}

void setup2D() {
    glClearColor(0,0,0,0);
    glMatrixMode(GL_PROJECTION);
    glLoadIdentity();
    gluOrtho2D(-1,1,-1,1);
}

void triangleWindow() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Original Triangle
    glColor3f(1, 1, 1);
    glBegin(GL_TRIANGLES);
    glVertex2f(0.0f, 0.3f);
    glVertex2f(-0.3f, -0.3f);
    glVertex2f(0.3f, -0.3f);
    glEnd();

    // Transformed Triangle
    glPushMatrix();
    glTranslatef(0.5f, 0.3f, 1.0f);
    glScalef(0.9f, 0.5f, 1.0f);
    glColor3f(0, 1, 0);

    glBegin(GL_TRIANGLES);
    glVertex2f(0.0f, 0.3f);
    glVertex2f(-0.3f, -0.3f);
    glVertex2f(0.3f, -0.3f);
    glEnd();

    glPopMatrix();

    drawText("Triangle", 0, 1, 0);
    glFlush();
}

void polygonWindow() {
    glClear(GL_COLOR_BUFFER_BIT);

    // Original Polygon
    glColor3f(1, 1, 1);
    glBegin(GL_POLYGON);
    glVertex2f(0.0f, 0.4f);
    glVertex2f(0.4f, 0.1f);
    glVertex2f(0.3f, -0.4f);
    glVertex2f(-0.3f, -0.4f);
    glVertex2f(-0.4f, 0.1f);
    glEnd();

    // Transformed Polygon

```

```

    glPushMatrix();
    glTranslatef(0.5f, 0.4f, 1.0f);
    glScalef(0.5f, 0.5f, 1.0f);
    glColor3f(1, 0, 1);

    glBegin(GL_POLYGON);
    glVertex2f(0.0f, 0.4f);
    glVertex2f(0.4f, 0.1f);
    glVertex2f(0.3f, -0.4f);
    glVertex2f(-0.3f, -0.4f);
    glVertex2f(-0.4f, 0.1f);
    glEnd();

    glPopMatrix();

    drawText("Polygon", 1, 0, 1);
    glFlush();
}
int main(int argc, char** argv) {
    glutInit(&argc, argv);
    glutInitDisplayMode(GLUT_SINGLE | GLUT_RGB);
    glutInitWindowSize(400, 400);

    glutCreateWindow("Translation 2D");
    setup2D();
    glutDisplayFunc(translate2D);

    glutCreateWindow("Scaling 2D");
    setup2D();
    glutDisplayFunc(scale2D);

    glutCreateWindow("Triangle 2D");
    setup2D();
    glutDisplayFunc(triangleWindow);

    glutCreateWindow("Polygon 2D");
    setup2D();
    glutDisplayFunc(polygonWindow);

    glutMainLoop();
    return 0;
}

```

# Output of Lab Task

